```matlab
clc;
clear;
close all;

% EX NO: 1
% Butterworth IIR High-Pass Filter Design and Analysis
% Date:

% Filter Specifications
Fs  = 10000;    % Sampling Frequency (Hz)
Fp  = 3000;     % Passband Frequency (Hz)
Fst = 2000;     % Stopband Frequency (Hz)
Rp  = 1;        % Passband Ripple (dB)
Rs  = 60;       % Stopband Attenuation (dB)

% Calculate Minimum Filter Order
[N, Wn] = buttord(Fp/(Fs/2), Fst/(Fs/2), Rp, Rs);

% Design Butterworth High-Pass Filter
[b, a] = butter(N, Wn, 'high');

% Magnitude and Phase Response
figure;
freqz(b, a, 1024, Fs);
title('Magnitude and Phase Response');

% Impulse Response
figure;
```

```
impz(b, a, 50, Fs);
grid on;
title('Impulse Response');

% Step Response
figure;
stepz(b, a);
grid on;
title('Step Response');

% Group Delay
figure;
grpdelay(b, a, 1024, Fs);
grid on;
title('Group Delay');

% Pole-Zero Plot
figure;
zplane(b, a);
grid on;
title('Pole-Zero Plot');

% Display Results
fprintf('Minimum Filter Order (N) = %d\n', N);
fprintf('Normalized Cutoff Frequency (Wn) = %.4f\n', Wn);
fprintf('Actual Cutoff Frequency = %.2f Hz\n', Wn * (Fs/2));
```

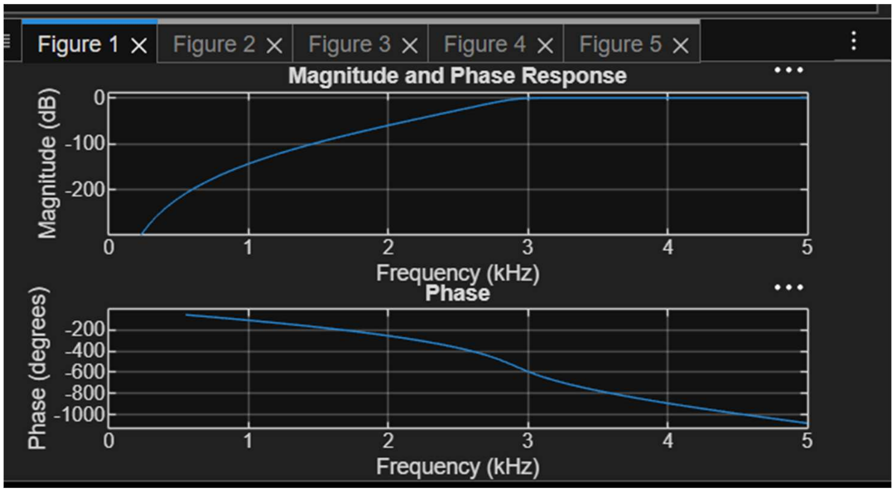


Figure 1 Figure 2 Figure 3 Figure 4 Figure 5
Magnitude and Phase Response
Magnitude (dB)
0
-100
-200
0 1 2 3 4 5
Frequency (kHz)
Phase
Phase (degrees)
-200
-400
-600
-800
-1000
0 1 2 3 4 5
Frequency (kHz)

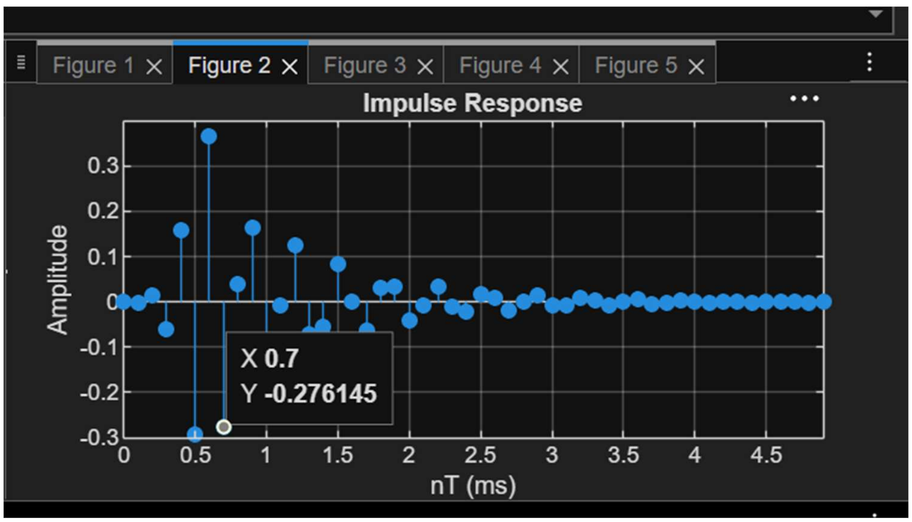


Figure 1 Figure 2 Figure 3 Figure 4 Figure 5
Impulse Response
Amplitude
0.3
0.2
0.1
0
-0.1
-0.2
-0.3
X 0.7
Y -0.276145
0 0.5 1 1.5 2 2.5 3 3.5 4 4.5
nT (ms)

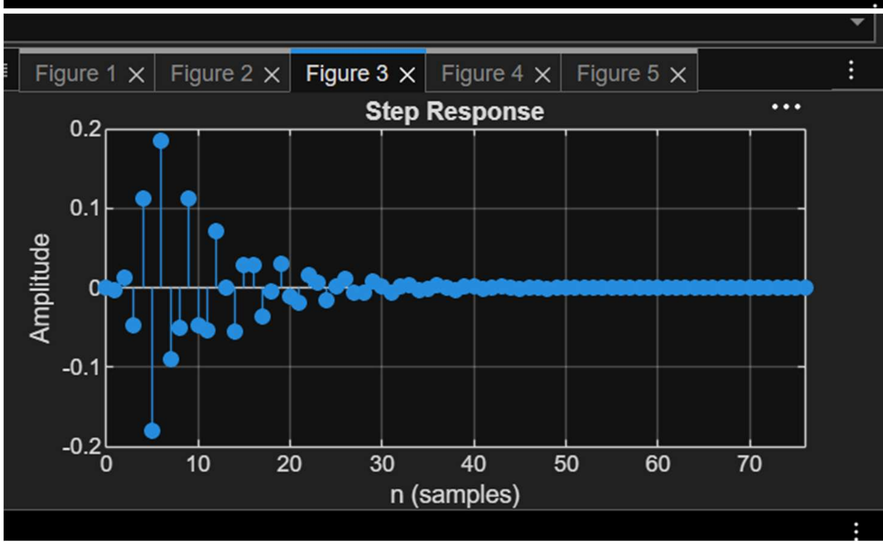


Figure 1 Figure 2 Figure 3 Figure 4 Figure 5
Step Response
Amplitude
0.2
0.1
0
-0.1
-0.2
0 10 20 30 40 50 60 70
n (samples)

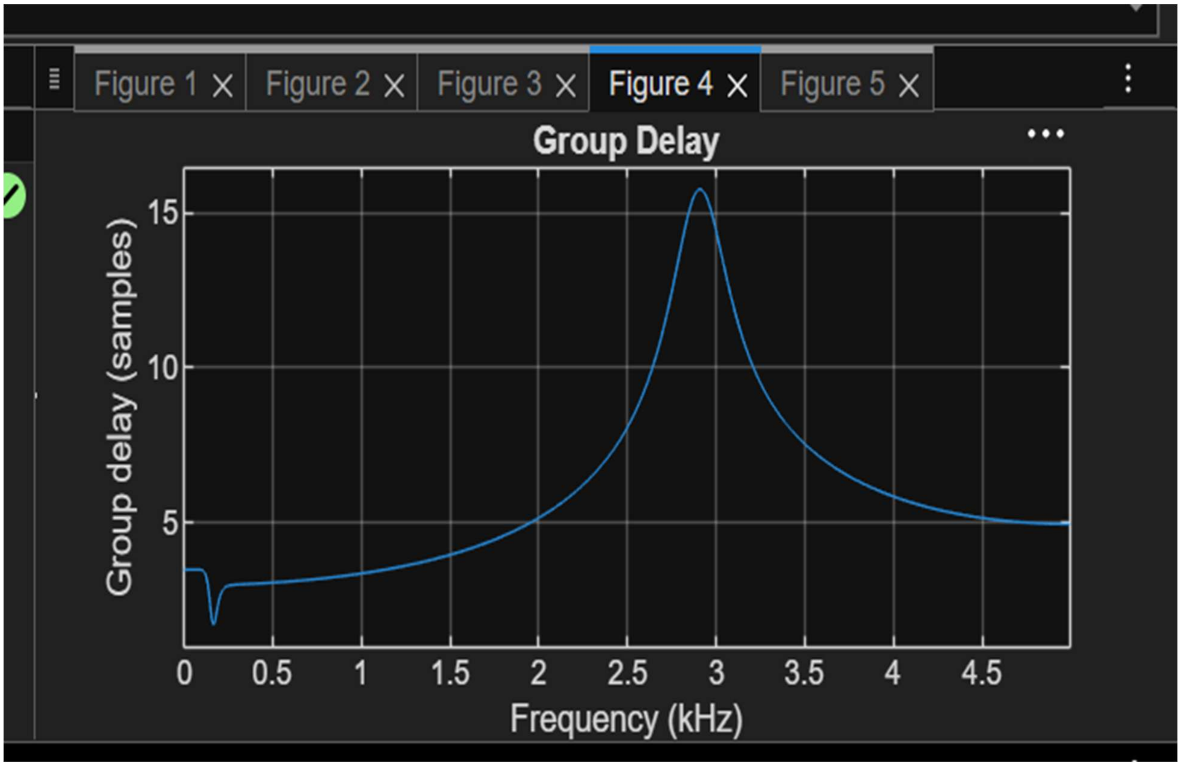
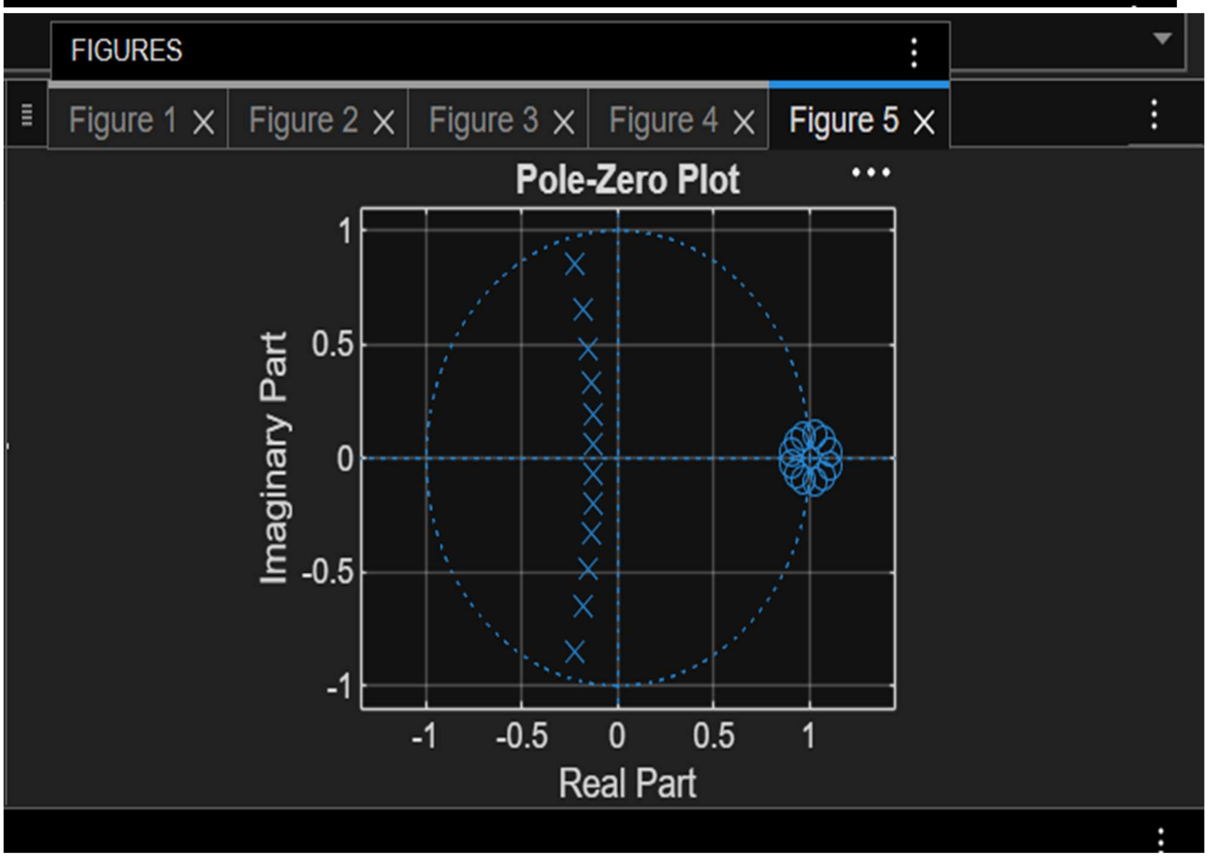


Figure 1
Figure 2
Figure 3
Figure 4
Figure 5
Group Delay
Group delay (samples)
15
10
5
0
0.5
1
1.5
2
2.5
3
3.5
4
4.5
Frequency (kHz)


FIGURES
Figure 1
Figure 2
Figure 3
Figure 4
Figure 5
Pole-Zero Plot
Imaginary Part
1
0.5
0
-0.5
-1
-1
-0.5
0
0.5
1
Real Part